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On the Spectrum of Evolution Operators Generated by Hyperbolic Systems

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We prove that the asymptotic behavior of semigroups generated by mixed-initial value problems for hyperbolic systems in one space variable is determined by the zeroes of the characteristic equation. The proof is based on the following reduction principle: we construct a reduced system (which is much simpler than the given one) such that the difference of the semigroups is a compact operator. The construction of the reduced system is presented in the non-autonomous case because it has implifications in the study of systems which are time periodic. © 1986 Academic Press, Inc.

I. INTRODUCTION

It is very well known that the asymptotic behavior of a C_0 semigroup $T(t)$, $t \geqslant 0$, in a Banach space X is determined by its spectrum $\sigma(T(t))$. For instance, to say that $T(t)$ decays to zero exponentially in an invariant subspace of finite codimension is the same as to say that the essential spectral radius of $T(t)$ (see definition in Sect. II) is less than one (this situation is particularly important in the analysis of the local behavior of the solutions of a semilinear equations in the neighborhood of an equilibrium point). However, in the applications all we know is the infinitesimal generator A and so it is highly desirable to obtain informations about $\sigma(T(t))$ from the knowledge of $\sigma(A)$ only. For semigroups which are either compact or analytic $\sigma(A)$ gives all the important about $\sigma(T(t))$ (that takes care of

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retarded equations and parabolic systems) but, in general, all we can say is that the point spectrum and the residual spectrum of $T(t)$ are exponentially related to those of A (see [1, 2]). For the continuum spectrum things are very different; in fact, a remarkable example in [1, 23.16], shows the existence of a group whose generator has empty spectrum, compact operator resolvent and, yet, the (continuum) spectrum of $T(t)$ is the circle with radius $e^{\pi|t|}$; in particular, $|T(t)|$ grows exponentially. This example shows that in general the continuum spectrum (hence the spectrum) of $T(t)$ cannot be detected by the inspection of the spectrum of A and the problem is to show that such a pathological behavior does not occur in "concrete" situations (which do not fit in those of analytic or compact semigroups). For difference and differential-difference equations that analysis has been carried out by Henry [3] who has shown, basically, that the asymptotic behavior of the semigroups generated by those equations is determined by the spectrum of the corresponding infinitesimal generator. In this paper we do the same thing for semigroups generated by hyperbolic systems in one space variable in L_p spaces, $1 \leqslant p < \infty$. The following standard procedure is adopted: if $T(t)$ and $T_0(t)$ are semigroups whose difference $T(t) - T_0(t)$ is compact and the asymptotic behavior of $T_0(t)$ is determined by the spectrum of its infinitesimal generator then the same thing happens to $T(t)$ (part e of Theorem II.B is the precise statement of this principle). This approach is similar (but slightly different) to the one present by Vidav [4]. In order to be able to use this procedure in our problem we construct such a $T_0(t)$ for which Henry's method applies. We present that reduction for the nonautonomous system because it has applications in the perturbation theory for linear systems with periodic coefficients. In periodicity in x is taken as boundary conditions and $p = 2$ then our results are similar to the ones obtained by Rauch and Taylor [5] by the Fourier method.

II. Statement of the Problem and the Main Results

We will be dealing with the following linear homogeneous hyperbolic system in one space variable in normal form:

$$\frac{\partial}{\partial t}\begin{pmatrix} u(t,x) \\ v(t,x) \end{pmatrix} + K(t,x)\frac{\partial}{\partial x}\begin{pmatrix} u(t,x) \\ v(t,x) \end{pmatrix} + C(t,x)\begin{pmatrix} u(t,x) \\ v(t,x) \end{pmatrix} = 0, \qquad 0 < x < l, \tag{II.1}$$

$$\frac{d}{dt}[v(t,l) - D(t)\,u(t,l)] = F(t)\,u(t,l) + G(t)\,v(t,l)$$

$$\text{with} \quad BC\colon u(t,0) = E(t)\,v(t,0)$$

where

(i) $K(t, x) = \operatorname{diag}[\lambda_1(t, x), \ldots, \lambda_N(t, x), \mu_{N+1}(t, x), \ldots, \mu_n(t, x)]$ is a diagonal $n \times n$ matrix whose entries are real values C^1 functions in t and x for (t, x) in $J \times [0, l]$, $J = [a, b]$ being a compact interval in $\mathbb{R}$, with $\lambda_i(t, x) > 0$, $i = 1, \ldots, N$, and $\mu_j(t, x) < 0$ for $j = N+1, \ldots, n$;

(ii) $C(t, x)$ is an $n \times n$ matrix whose entries $c_{ij}(t, x)$ are real (or complex) values continuous functions of (t, x) in $J \times [0, l]$ and either C^1 in t or C^1 in x;

(iii) $u(t, x) = \operatorname{col}[u_i(t, x)]$, $i = 1, \ldots, N$ is a column vector in $\mathbb{R}^N$ (or $\mathbb{C}^N$) and $v(t, x) = \operatorname{col}[v_j(t, x)]$, $j = N+1, \ldots, n$ is a column vector in $\mathbb{R}^{n-N}$ (or $\mathbb{C}^{n-N}$);

(iv) $D(t)$, $E(t)$, $F(t)$, and $G(t)$ are matrices of appropriate sizes whose entries are real (or complex) valued C^1 functions defined on J.

We also assume that $K(t, x)$ and $C(t, x)$ satisfy the following condition:

(v) if $k \neq m$ and $\lambda_k(t, x) = \lambda_m(t, x)$ (or $\mu_k(t, x) = \mu_m t, x)$) somewhere in $J \times [0, l]$ then $c_{km}(t, x)$ vanishes identically on $J \times [0, l]$.

Assumption (v) plays an important role in our theory and it is satisfied in all applications we have encountered.

In fact, systems of type (II.1) governing physical problems are of the form

$$\frac{\partial w}{\partial t} + M(t, x)\frac{\partial w}{\partial x} + N(t, x)w = 0$$

plus boundary conditions, where $M(t, x)$ and $N(t, x)$ are block diagonal where the blocks $M_i(t, x)$ and $N_i(t, x)$ are 2×2 matrices with $M_i(t, x) = \operatorname{diag}(\varepsilon_i(t, x), -\varepsilon_i(t, x))$, $\varepsilon_i(t, x) > 0$. It is easy to see that those systems satisfy assumption (v).

Remark. The conditions at $x = 0$ and $x = l$ can be combined in several ways giving rise to other problems which can be treated in a similar way. If $K(t, x)$ and $C(t, x)$ are 2π-periodic in x then we can take 2π-periodicity in x as a boundary condition for the solutions and this problem has also a similar treatment.

For any p, $1 \leqslant p < \infty$ we consider the Banach space $X_p = (L_p([0, l]))^n \times \mathbb{R}^{n-N}$ (or $(L_p([0, l]))^n \times \mathbb{C}^{n-N}$), where $L_p([0, l])$ denotes the set of the real (or complex) valued functions whose power p is integrable on $[0, l]$, and we define

$$A(t)\colon \mathscr{D}(A(t)) \subset X_p \to X_p$$

by

$$A(t)(u, v, d) = \left[-K(t, x)\frac{\partial}{\partial x}\begin{pmatrix} u \\ v \end{pmatrix} - C(t, x)\begin{pmatrix} u \\ v \end{pmatrix}, F(t)\,u(l) + G(t)\,v(l) \right]$$

where

$$\mathscr{D}(A(t)) = \{(u, v, d) \in X_p : u \in (H_{1,p}([0, l]))^N,\ v \in (H_{1,p}([0, l]))^{n-N},$$
$$u(0) = E(t)\, v(0),\ d = v(l) - D(t)\, u(l)\},$$

$H_{1,p}[0, l]$ denoting the usual Sobolev space, With this setting system (II.1) can be viewed as an abstract equation $\dot{w} = A(t)w$ in X_p and, under assumptions (i)–(iv), $A(t)$ generates an evolution operator $T(t, s)$, $s \leqslant t$, in the sense of Krein [6]. If E and D are independent of t and $C(t, x)$ is C^1 in t then the existence of $T(t, s)$ follows from the classical theory of Krein [6] because defining an equivalent norm in X_p by

$$\|(u, v, d)\|^p = \sum_{i=1}^{N} \int_0^l p_i(x) |u_i(x)|^p \, dx + \sum_{j=1}^{n} q_j(x) |v_j(x)|^p \, dx + |d|^p$$

and choosing $p_i(x)$ and $q_j(x)$ properly, it can be shown that

$$j(u, v, d)(A(t)(u, v, d)) \leqslant m \|(u, v, d)\|^2$$

(or Re $j(u, v, d)(A(t)(u, v, d)) \leqslant m \|(u, v, d)\|^2$ in the complex case) where m is a constant and j denotes the duality map; if $E(t)$ and $D(t)$ are C^2 then the existence of $T(t, s)$ follows from Kato's theory [7] and the general case can be proved by an iteration method similar to the one presented in Courant and Hilbert [8].

Together with system (II.1) we consider the following three reduced systems:

$$\frac{\partial}{\partial t}\begin{pmatrix} u(t, x) \\ v(t, x) \end{pmatrix} + K(t, x) \frac{\partial}{\partial x}\begin{pmatrix} u(t, x) \\ v(t; x) \end{pmatrix}$$
$$+ C_0(t, x)\begin{pmatrix} u(t, x) \\ v(t, x) \end{pmatrix} = 0, \qquad 0 < x < l, \tag{II.2}$$
$$\frac{d}{dt}[v(t, l) - D(t)\, u(t, l)] = F(t)\, u(t, l) + G(t)\, v(t, l)$$

with boundary condition: $u(t, 0) = E(t)\, v(t, 0)$;

$$\frac{\partial}{\partial t}\begin{pmatrix} u(t, x) \\ v(t, x) \end{pmatrix} + K(t, x) \frac{\partial}{\partial x}\begin{pmatrix} u(t, x) \\ v(t, x) \end{pmatrix}$$
$$+ C_0(t, x)\begin{pmatrix} u(t, x) \\ v(t, x) \end{pmatrix} = 0, \qquad 0 < x < l, \tag{II.3}$$
$$\frac{d}{dt}[v(t, l) - D(t)\, u(t, l)] = 0$$

with boundary condition: $u(t, 0) = E(t)\, v(t, 0)$

and

$$\frac{\partial}{\partial t}\begin{pmatrix} u(t,x) \\ v(t,x) \end{pmatrix} + K(t,x)\frac{\partial}{\partial x}\begin{pmatrix} u(t,x) \\ v(t,x) \end{pmatrix} + C_0(t,x)\begin{pmatrix} u(t,x) \\ v(t,x) \end{pmatrix} = 0, \qquad 0 < x < l, \tag{II.4}$$

with boundary conditions:

$$u(t,0) = E(t)\,v(t,0) \text{ and } v(t,l) = D(t)\,u(t,l).$$

where $C_0(t,x) = \operatorname{diag}(c_{11}(t,x),\ldots, c_{nn}(t,x))$.

Each system can be viewed as an abstract equation $\dot{w} = A_i(t)w$, $i = 2, 3, 4$, with $A_i(t)$ and $\mathscr{D}(A_i(t))$ defined in the obvious way and the corresponding evolution operators will be denoted by $T_i(t,s)$, $s \leqslant t$, $i = 2, 3, 4$. For systems (II.2) and (II.3) the phase space is also X_p, but for system (II.4) the phase space is $(L_p[0,l])^n$. Letting $X_p^0 = \{(u,v,d) \in X_p : d = 0\}$ it is easy to see that X_p^0 is invariant under $T_3(t,s)$ and $T_3(t,s) = T_4(t,s)$ on X_p^0 (we are identifying $(u,v,0)$ with (u,v)); we denote by $\pi: X_p \to X_p$ the projection onto X_p^0 defined by $\pi(u,v,d) = (u,v,0)$.

Before stating our main results we need to define a few concepts.

DEFINITION. A complex number λ is a normal point of a bounded linear operator T in a Banach space X if either λ belongs to the resolvent set of T or λ is an isolated point in the spectrum of T and the corresponding spectral projection has finite dimensional range.

DEFINITION. The essential spectral radius $r_{\text{ess}}(T)$ of a bounded linear operator T is the infimum of the set of the numbers $r > 0$ such that $\sigma(T) \cap \{\lambda : |\lambda| > r\}$ consists of normal points of T; in other words, $r_{\text{ess}}(T)$ is the radius of the complement of the normal points.

It is very well known (see [9], for instance) that two bounded operators whose difference is compact have the same essential spectral radius.

We are ready to state our main theorems.

THEOREM A. *Under assumptions* (i) *to* (v), *for each* (t,s), $s \leqslant t$, *the difference*: $T(t,s) - T_4(t,s)\Pi: X_p \to X_p$ *is a compact operator* (*in fact*, $\{T(t,s) - T_4(t,s)\pi)w,\ s \leqslant t,\ s,\ t \text{ in } J,\ |w| \leqslant 1\}$ *is precompact*). *In particular*, $r_{\text{ess}}(T(t,s)) = r_{\text{ess}}(T_4(t,s))$.

THEOREM B. *If the matrices* K, C, D, E, F, *and* G *are independent of* t (*in this case the operators previously defined generate strongly continuous semigroups*) *then there are two entire functions* $h(\lambda)$ *and* $h_4(\lambda)$ (*which will be defined later in the proof*), $h_4(\lambda)$ *being a finite exponential sum such that*:

(a) $\sigma(A) = \{\lambda \in \mathbb{C}: h(\lambda) = 0\}$;

(b) $\sigma(A_4) = \{\lambda \in \mathbb{C}: h_4(\lambda) = 0\}$;

(c) *the resolvent operators* $R(\lambda, A)$ *and* $R(\lambda, A_4)$ *are compact (whenever they are defined)*;

(d) *if* α_0 *is defined by* $\alpha_0 = \sup\{\operatorname{Re}\lambda: h_4(\lambda) = 0\}$ *then for any* $\alpha > \alpha_0$, $\sigma(A) \cap \{\lambda: \operatorname{Re}\lambda \geq \alpha\}$ *is finite*;

(e) *if* $|z| > e^{\alpha_0 t}$ *then* z *belongs to the spectrum of* $T(t)$ *if and only if* $z = e^{\lambda t}$ *for some* λ *satisfying* $h(\lambda) = 0$.

(f) (*exponential dichotomy*) *if* $\alpha > \alpha_0$ *and there is no solution of* $h(\lambda) = 0$ *satisfying* $\operatorname{Re}\lambda = \alpha$ *then* $|T(t)(I-P)| \leq Ke^{\alpha t}$, $t \geq 0$, *where* K *is a constant and* P *is the finite dimensional spectral projection of* A *corresponding to the roots of* $h(\lambda) = 0$ *lying on the half plane* $\operatorname{Re}\lambda > \alpha$.

Remarks. (i) Assertions (a), (e), and (f) in Theorem B tell us that the asymptotic behavior of $T(t)$ can be determined once $\sigma(A)$ is known.

(ii) If P is defined as in part (f) of Theorem B then $P(X_p)$ and $(I-P)(X_p)$ are invariant under $T(t)$, $t \geq 0$; since $P(X_p)$ is finite dimensional the restriction of $T(t)$ to $P(X_p)$ is the exponential of a "matrix" whose eigenvalues are the roots of $h(\lambda) = 0$ satisfying $\operatorname{Re}\lambda > \alpha$; it can be shown [10] that the dimension of the range of the spectral projection of A corresponding to a single characteristic root is equal to its multiplicity as a root of $h(\lambda) = 0$, and then the dimension of $P(X_p)$ is the sum of the multiplicities of the characteristic roots satisfying $\operatorname{Re}\lambda > \alpha$.

(iii) The functions $h(\lambda)$ and $h_4(\lambda)$ are independent of p; in particular, the essential spectral radius of $T(t)$ is independent of p. If the coefficients of system (II.1) are, say, periodic in t with period $\tau > 0$ then the essential spectral radius of $T(\tau, 0)$ may depend on p (see [11]).

(iv) The finite exponential sum $h_4(\lambda)$ can be constructed explicitly in terms of E, D, and the integrals on $[0, l]$ of the elements of $K(x)$ and $C_0(x)$; however, the construction of $h(\lambda)$ is difficult even in the case of constant coefficients. Zeroes of exponential sums have been studied in [12, 13, and 14], among others.

III. Proofs of the Main Results

The proofs will be broken in several lemmas. Lemmas 1, 2, and 3 deal with the evolution operators $T(t, s)$, $\tilde{T}(t, s)$ in a Banach space X, $s \leq t$, s, t in some interval $J = [a, b]$, generated by the evolution equations $\dot{u} = A(t)u$, $\dot{u} = \tilde{A}(t)u$, respectively, where $A(t) = \tilde{A}(t) + B(t)$, $B(t)$ being bounded and strongly continuous for t in J; $J_0 = [a_0, b_0]$ will denote a compact subinterval of J.

LEMMA 1. *If* $\int_{a_0}^{t} \tilde{T}(t, s)\, B(s)\, \tilde{T}(s, a_0)\, ds$ *is a compact operator for any* t *in* J_0 *then* $\{\int_{a_0}^{t} \tilde{T}(t, s)\, B(s)\, \tilde{T}(s, a_0)\, u\, ds,\ t \in J_0,\ |u| \leqslant 1\}$ *is precompact.*

Proof. Given $\varepsilon > 0$ let $n_0 = n_0(\varepsilon)$ be an integer such that

$$\left| \int_{t}^{t+r} \tilde{T}(t+r, s)\, B(s)\, \tilde{T}(s, a_0)\, ds \right| < \varepsilon$$

for any t in J_0 and any $r \geqslant 0$ satisfying $r < (b_0 - a_0)/n_0$ and let t_n be defined by $t_n = a_0 + n((b_0 - a_0)/n_0)$, $n = 0, 1, \ldots, n_0 - 1$; for t belonging to $[t_n, t_{n+1}]$ and $r = t - t_n$ we have

$$\begin{aligned}
&\int_{a_0}^{t} \tilde{T}(t, s)\, B(s)\, \tilde{T}(s, a_0)\, ds \\
&\quad = \int_{a_0}^{t_n + r} \tilde{T}(t_n + r, s)\, B(s)\, \tilde{T}(s, a_0)\, ds \\
&\quad = \tilde{T}(t_n + r, t_n) \int_{a_0}^{t_n} \tilde{T}(t_n, s)\, B(s)\, \tilde{T}(s, a_0)\, ds \\
&\qquad + \int_{t_n}^{t_n + r} \tilde{T}(t_n + r, s)\, B(s)\, \tilde{T}(s, a_0)\, ds.
\end{aligned}$$

Since $\{\int_{a_0}^{t_n} \tilde{T}(t_n, s)\, B(s)\, \tilde{T}(s, a_0)\, u\, ds,\ |u| \leqslant 1\}$, $n = 0, 1, \ldots, n_0 - 1$ are precompact and $(t, s, v) \to \tilde{T}(t, s)v$, $s \leqslant t$, is a continuous function it follows that

$$K_n = \left\{ \tilde{T}(t_n + r, t_n) \int_{a_0}^{t_n} \tilde{T}(t_n, s)\, B(s)\, \tilde{T}(s, a_0) u\, ds,\ r \in \left[0, \frac{b_0 - a_0}{n_0}\right],\ |u| \leqslant 1 \right\}, \qquad n = 0, \ldots, n_0 - 1,$$

are precompact as well as $K_\varepsilon = K_0 \cup \cdots \cup K_{n_0 - 1}$ and this proves the lemma.

LEMMA 2. *Suppose there is a* $\delta > 0$ *such that* $T(t, a_0) - \tilde{T}(t, a_0)$ *is compact for any* t, a_0, b_0 *in* J *such that* $a_0 \leqslant t \leqslant b_0 \leqslant a_0 + \delta$; *then* $T(t, a) - \tilde{T}(t, a)$ *is compact for any* t *in* J. *Furthermore, if* $\{(T(t, a_0) - \tilde{T}(t, a_0))(u),\ |u| \leqslant 1,\ t \in J_0\}$ *is precompact for any* a_0, b_0 *in* J *such that* $b_0 \leqslant a_0 + \delta$ *then*

$$\{(T(t, a) - \tilde{T}(t, a))(u),\ |u| \leqslant 1,\ t \in J\}$$

is precompact.

Proof. If $a \leqslant t \leqslant a + \delta$, the assertion follows from the assumption; if $a + \delta \leqslant t \leqslant a + 2\delta$, defining $c = t - (a + \delta)$, we have $0 \leqslant c \leqslant \delta$ and then the equality

$$
\begin{aligned}
&T(a+\delta+c, a)-\tilde{T}(a+\delta+c, a)\\
&\quad=T(a+\delta+c, a+\delta)\, T(a+\delta, a)-\tilde{T}(a+\delta+c, a+\delta)\, \tilde{T}(a+\delta, a)\\
&\quad=[T(a+\delta+c, a+\delta)-\tilde{T}(a+\delta+c, a+\delta)]\, T(a+\delta, a)\\
&\qquad+\tilde{T}(a+\delta+c, a+\delta)[T(a+\delta, a)-\tilde{T}(a+\delta, a)]
\end{aligned}
$$

proves that the assertion holds on $[a, a+2\delta]$. Thus the lemma follows an inductive argument.

LEMMA 3. *Suppose there is a* $\delta>0$ *such that* $\int_{a_0}^{t} \tilde{T}(t, s)\, B(s)\, \tilde{T}(s, a_0)\, ds$ *is a compact operator for any* a_0, t *in* J *such that* $a \leqslant a_0 \leqslant t \leqslant a_0+\delta$; *then* $T(t, a)-\tilde{T}(t, a)$ *is a compact operator for any* t *in* J; *(in fact,* $\{(T(t, a)-\tilde{T}(t, a))u,\ |u| \leqslant 1,\ t \in J\}$ *is precompact.*

Proof. According to Lemma 2 it is enough to prove that

$$\{(T(t, a_0)-\tilde{T}(t, a_0))u,\ |u| \leqslant 1,\ t \in J_0\}$$

is precompact for any t and a_0 in J such that $a_0 \leqslant t \leqslant b_0=a_0+\delta$. Since $T(t, a_0)=\tilde{T}(t, a_0)+\int_{a_0}^{t} \tilde{T}(t, s)\, B(s)\, T(s, a_0)\, ds$, we conclude that $U(t)=T(t, a_0)-\tilde{T}(t, a_0)$ satisfies the Volterra integral equation

$$U(t)=\int_{a_0}^{t} \tilde{T}(t, s)\, B(s)\, \tilde{T}(s, a_0)\, ds+\int_{a_0}^{t} \tilde{T}(t, s)\, B(s)\, U(s)\, ds.$$

Let $C(J_0, X)$ be the space of continuous from J_0 into X with the sup norm. First of all we claim that

$$\left\{\int_{a_0}^{\cdot} \tilde{T}(\cdot, s)\, B(s)\, \tilde{T}(s, a_0) u\, ds,\ |u| \leqslant 1\right\}$$

is precompact in $C(J_0, X)$; in fact, from Lemma 1 we know that

$$\left\{\int_{a_0}^{t} \tilde{T}(t, s)\, B(s)\, \tilde{T}(s, a_0) u\, ds,\ |u| \leqslant 1,\ t \in J_0\right\}$$

is precompact; furthermore for any $h>0$ we have

$$
\begin{aligned}
&\left|\int_{a_0}^{t+h} \tilde{T}(t+h, s)\, B(s)\, \tilde{T}(s, a_0) u\, ds-\int_{a_0}^{t} \tilde{T}(t, s)\, B(s)\, \tilde{T}(a, a_0) u\, ds\right|\\
&\quad\leqslant\left|\int_{t}^{t+h} \tilde{T}(t+h, s)\, B(s)\, \tilde{T}(s, a_0) u\, ds\right|\\
&\qquad+\left|(\tilde{T}(t+h, t)-I)\int_{a_0}^{t} \tilde{T}(t, s)\, B(s)\, \tilde{T}(s, a_0) u\, ds\right|
\end{aligned}
$$

and since these expressions go to zero as h goes to zero *uniformly* for t in J_0 and u in the unit ball the claim follows from Arzela–Ascoli theorem. Next we notice that the operator $\mathscr{K}: C(J_0, X) \to C(J_0, X)$ that takes $p \in C(J_0, X)$ into the unique solution $y(t)$ of $y(t) = p(t) + \int_{a_0}^{t} \tilde{T}(t, s)\, B(s)\, y(s)\, ds$ is continuous and this proves the lemma because

$$\{U(\cdot)u, |u| \leqslant 1\} = \mathscr{K}\left\{\int_{a_0}^{\cdot} \tilde{T}(\cdot, s)\, B(s)\, \tilde{T}(s, a_0)u\, ds, |u| \leqslant 1\right\}.$$

The next lemma deals with criteria of precompactness of subsets of $L_p([0, l], \mathbb{R})$ (or $L_p([0, l], \mathbb{C})$), $1 \leqslant p < \infty$.

LEMMA 4. (i) *Let $\bar{l}$ be a point in $[0, l]$ and H be a subset of $L_p([0, l])$ such that the subsets of $L_p([0, \bar{l}])$ and $L_p([\bar{l}, l])$ obtained by restricting the elements of H are precompact; then H is precompact;*

(ii) *if $K: [c, d] \times [a, b] \to \mathbb{R}$ (or $\mathbb{C}$) and*

$$r_1, r_2: [c, d] \to [a, b]$$

are continuous functions then the operator $\mathscr{K}_1: L_p([a, b]) \to L_p([c, d])$ defined by

$$(\mathscr{K}_1 f)(x) = \int_{r_1(x)}^{r_2(x)} K(x, s)\, f(s)\, ds \qquad \text{is compact;}$$

(iii) *If $I \subset \mathbb{R}$ is compact interval and $\alpha: I \times [c, d] \to [a, b]$ is continuous in $(s, x) \in I \times [c, d]$, C^1 in s with $(\partial\alpha/\partial s)(s, x) \neq 0$ everywhere, $r_1, r_2: [c, d] \to [a, b]$ and $K: [c, d] \times I \to \mathbb{R}$ (or $\mathbb{C}$) are continuous then $\mathscr{K}_2: L_p([a, b]) \to L_p([c, d])$ defined by*

$$(\mathscr{K}_2 f)(x) = \int_{r_1(x)}^{r_2(x)} K(x, s)\, f(\alpha(s, x))\, ds \qquad \textit{is compact.}$$

Proof. (i) and (ii) are obvious; (iii) follows from (ii) because if we make the change of variable $\alpha(s, x) = \tau$ and denote by $\beta(\tau, x)$ its inverse $s = \beta(t, x)$ then

$$(\mathscr{K}_2 f)(x) = \int_{\alpha(r_1(x), x)}^{\alpha(r_2(x), x)} K(x, \beta(\tau, x))\, \frac{\partial \beta}{\partial \tau}(\tau, x)\, f(\tau)\, d\tau$$

and the lemma is proved.

Remark. For $\alpha(s, x) \equiv x$ the operator $\mathscr{K}_2$ defined above is the multiplication by a fixed function and then it is not compact; of course the condition $\partial\alpha/\partial s \neq 0$ is not satisfied.

Proof of Theorem A. We start by showing that $T(t, a) - T_3(t, a)$ is compact. Notice that Lemma 2 cannot be used because the operator

$(u, v, d) \in X_p \to F(t)\, u(l) + G(t)\, v(l)$ is unbounded, and so we are going to study the operators $T(t, a) - T_2(t, a)$ and $T_2(t, a) - T_3(t, a)$. In order to accomplish that we start by writing down the explicit representation of $T_2(t, a)$ and $T_3(t, a)$; that explicit representation can be obtained by the method of the characteristics and "reflections" of them at the boundary because the coupling among the equations of systems (II.2) and (II.3) takes place at the endpoints only. Let $\varphi_i(t, s; x)$, $i = 1,..., N$ and $\psi_j(t, s; x)$, $j = N+1,..., n$ be the solutions of $dx/dt = \lambda_1(t, x)$ and $dx/dt = \mu_j(t, x)$ satisfying $\varphi_i(s, s; x) = x$ and $\psi_j(s, s; x) = x$, respectively; for each (t, x) we define $\tau_i(t, x)$, $i = 1,..., N$ and $\tau_j(t, x)$, $j = N+1,..., n$ by $\varphi_i(\tau, t; x) = 0$ and $\psi_j(\tau, t; x) = l$, respectively. It is easy to see that there is a $\delta > 0$ such that if $a_0 \leqslant t \leqslant b_0 \leqslant a_0 + \delta$ then the representations of $T_2(t, a_0)$ and $T_3(t, a_0)$ can be obtained by means of at most one "reflection" of each characteristic (see Figs. 1 and 2). With that choice of δ, a_0, and b_0 an elementary computation shows that $T_2(t, a_0)(u_0, v_0, d_0) = (u(t, a_0), v(t, a_0), d(t, a_0))$, where

$$
\begin{aligned}
&u^i(t, a_0, x) \\
&\quad = \exp - \int_{\tau_i(t,x)}^{t} c_{ii}(s, \varphi_i(s, t; x))\, ds \\
&\qquad \times \Bigg[\sum_{j=N+1}^{n} e_{ij}(\tau_i(t, x)) \\
&\qquad \times \left(\exp - \int_{a_0}^{\tau_i(t,x)} c_{jj}(s, \psi_j(s, \tau_i(t, x); 0)\, ds \right) \\
&\qquad \times v_0^j(\psi_j(a_0, \tau_i(t, x); 0)) \Bigg]
\end{aligned}
$$

if $x \leqslant \varphi_i(t, a_0; 0)$ and

$$
u^i(t, a_0, x) = \left(\exp - \int_{a_0}^{t} c_{ii}(s, \varphi_i(s, t; x))\, ds \right) u_0^i(\varphi_i(a_0, t; x))
$$

$$
\text{if} \quad \varphi_i(t, a_0, 0) \leqslant x \leqslant l;
$$

$$
\begin{aligned}
v^j(t, a_0, x) &= \exp - \int_{\tau_j(t,x)}^{t} c_{jj}(s, \psi_j(s, t; x))\, ds \\
&\quad \times \Bigg[d^j(\tau_j(t, x)) + \sum_{k=1}^{N} d_{jk}(\tau_j(t, x)) \\
&\quad \times \left(\exp - \int_{a_0}^{\tau_j(t,x)} c_{kk}(s, \varphi_k(s, \tau_j(t, x); l))\, ds \right) u_0^k(\varphi_k(a_0, \tau_j(t, x); l)) \Bigg)
\end{aligned}
$$

$$
\text{if} \quad \psi_j(t, a_0; l) \leqslant x \leqslant l
$$

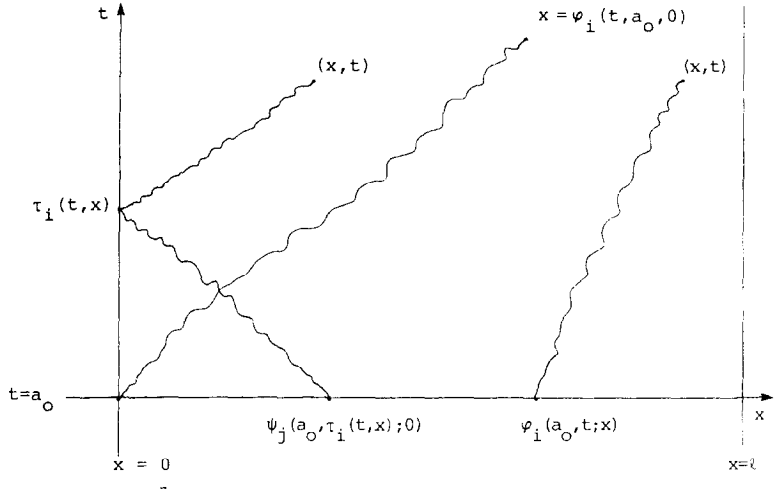


FIGURE 1

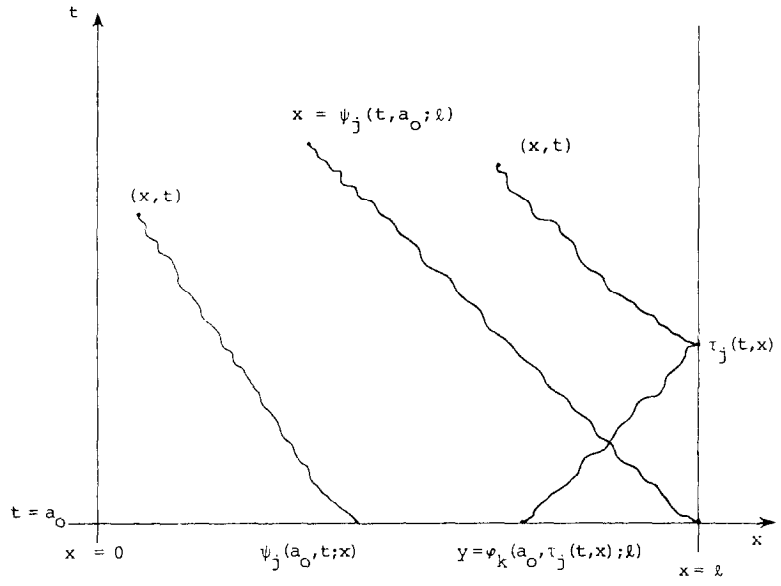


FIGURE 2

and

$$v^j(t, a_0, x) = \left(\exp - \int_{a_0}^{t} c_{jj}(s, \psi_j(s, t; x))\, ds\right) v_0^j(\psi_j(a_0, t; x))$$

$$\text{if} \quad 0 \leqslant x \leqslant \psi_j(t, a_0; l);$$

$$d(t) = X(t, a_0)\, d_0 + \int_{a_0}^{t} X(t, s)[G(s)\, D(s) + F(s)]\, u(s, a_0, l)\, ds,$$

$X(t, s)$ being the fundamental matrix of the linear system $\dot{d} = G(t)d$. Notice that $u(s, a_0, l)$ is given in terms of the initial conditions, and then the term $u(s, a_0, l)$ appearing in the expression of $d(t)$ is known and this expression has to be inserted in the representation of $v(t, a_0, x)$.

Since $T_3(t, a_0)$ is obtained from $T_2(t, a_0)$ by taking $F(t)$ and $G(t)$ identically zero we see that

$$(T_2(t, a_0) - T_3(t, a_0))(u_0, v_0, d_0) = (0, w(t, a_0), d(t, a_0) - d_0)$$

where

$$w^j(t, a_0, x) = \exp - \int_{\tau_j(t,x)}^{t} c_{jj}(s, \psi_j(s, t; x))\, ds (d^j(\tau_j(t, x)) - d_0)$$

$$\text{if} \quad \psi_j(t, a_0; l) \leqslant x \leqslant l$$

and

$$w^j(t, a_0, x) = 0 \qquad \text{if} \quad 0 \leqslant x \leqslant \psi_j(t, a_0; l)$$

and $d(t, a_0)$ is above. Taking in account that $\partial \tau_j / \partial x \neq 0$ (this derivative will be computed later in the proof), $\partial \varphi_i / \partial x \neq 0$ and $|(d/dt)\, d(t)|_{L_p[a_0, b_0]} \leqslant K(|d_0| + |u(t, a_0, l)_{L_p(a_0, b_0]})$, it is not difficult to see that

$$\{(T_2(t, s) - T_3(t, s))(u_0, v_0, d_0), a_0 \leqslant s \leqslant t \leqslant b_0, |(u_0, v_0, d_0)| \leqslant 1\}$$

is precompact.

Next we are going to use Lemma 2 to show that

$$\{(T(t, a_0) - T_2(t, a_0))(u_0, v_0, d_0), a_0 \leqslant t \leqslant b_0, |(u_0, v_0, d_0)| \leqslant 1\}$$

is precompact and so we have to verify that

$$\int_{a_0}^{t} T_2(t, s)\, B(s)\, T_2(s, a_0)\, ds$$

is a compact operator, where $B(s)(u, v, d) = (\tilde{C}(s, x)\binom{u(x)}{v(x)}, 0)$ with $\tilde{c}_{ij}(s, x) = c_{ij}(s, x)$ if $i \neq j$ and $\tilde{c}_{ii}(s, x) = 0$. From the equality

$$\begin{aligned} T_2(t, s)\, B(s)\, T_2(s, a_0) &= T_3(t, s)\, B(s)\, T_3(s, a_0) \\ &\quad + (T_2(t, s) - T_3(t, s))\, B(s)\, T_2(s, a_0) \\ &\quad + T_3(t, s)\, B(s)(T_2(s, a_0) - T_3(s, a_0)) \end{aligned}$$

and using the property of $T_2(t, s) - T_3(t, s)$ proved above we conclude it suffices to show

$$\int_{a_0}^{t} T_3(t, s)\, B(s)\, T_3(s, a_0)\, ds$$

is compact. The operator defined by the third component of this integral is compact because it has finite dimensional range and we will use Lemma 3 to prove that the first and the second components also define compact operators; we will deal with the first component only because the second has a similar treatment. We will use the notations

$$(\tilde{u}(s, a_0), \tilde{v}(s, a_0), \tilde{d}(s, a_0)) = B(s)\, T_3(s, a_0)(u_0, v_0, d_0)$$

and

$$\begin{aligned} (\bar{u}(t, s), \bar{v}(t, s), \bar{d}(t, s)) &= T_3(t, s)(\tilde{u}(s, a_0), \tilde{v}(s, a_0), \tilde{d}(s, a_0)) \\ &= T_3(t, s)\, B(s)\, T_3(s, a_0)(u_0, v_0, d_0). \end{aligned}$$

For $0 \leqslant x \leqslant \varphi_i(t, a_0; 0)$, we have

$$\int_{a_0}^{t} \bar{u}^i(t, s, x)\, ds = \int_{a_0}^{\tau_i(t,x)} \bar{u}^i(t, s, x)\, ds + \int_{\tau_i(t,x)}^{t} \bar{u}^i(t, s, x)\, ds;$$

moreover, for $a_0 \leqslant s \leqslant \tau_i(t, x)$, $\bar{u}^i(t, s, x)$ is given by

$$\bar{u}^i(t, s, x) = \sum_{j=N+1}^{n} \alpha_{ij}(t, s, x)\, \tilde{v}^j(s, a_0, \psi_j(s, \tau_i(t, x); 0))$$

where α_{ij} are continuous functions of their arguments and

$$\begin{aligned} \tilde{v}^j(s, a_0, \psi_j(s, \tau_i(t, x); 0)) = \tilde{v}^j(s, a_0, y) &= \sum_{k=1}^{N} \tilde{c}_{jk}(s, y)\, u^k(s, a_0, y) \\ &\quad + \sum_{k=N+1}^{n} \tilde{c}_{jk}(s, y)\, v^k(s, a_0, y) \end{aligned}$$

with $y = \psi_j(s, \tau_i(t, x); 0)$; we conclude $\int_{a_0}^{\tau_i(t,x)} \bar{u}^i(t, s, x)\, ds$ is the sum of terms of type

$$\int_{a_0}^{\tau_i(t,x)} \tilde{c}_{jk}(s, y)\, \alpha_{ij}(t, s, x)\, u^k(s, a_0, y)\, ds \tag{II.5}$$

and

$$\int_{a_0}^{\tau_i(t,x)} \tilde{c}_{jk}(s, y)\, \alpha_{ij}(t, s, x)\, v^k(s, a_0, y)\, ds. \tag{II.6}$$

In order to analyse the terms of type (II.5) we define $s_{ikj}(t, a_0, x)$ by the equation $\psi_i(s, \tau_i(t, x); 0) = \varphi_k(s, a_0; 0)$ and we break $\int_{a_0}^{\tau_i(t,x)}$ into the sum of $\int_{a_0}^{s_{ikj}(t,a_0,x)}$ and $\int_{s_{ikj}(t,a_0,x)}^{\tau_i(t,x)}$; the first integral can we written in the form

$$\int_{a_0}^{s_{ikj}(t,a_0,x)} \tilde{c}_{jk}(s, y)\, \alpha_{ijk}(t, s, x)\, u_{k0}^k(a_0, s; y))\, ds$$

and the second in the form

$$\int_{s_{ikj}(t,a_0,x)}^{\tau_i(t,x)} \tilde{c}_{jk}(s, y)\, \alpha_{ijk}(t, s, x)\, v_0^k(\psi_k(a_0, \tau_k(s, y); 0))\, ds$$

where α_{ijk} are continuous of their arguments. Next we show that the derivatives of the arguments of u_0^k and v_0^k with respect to s do not vanish. Recalling that $y = \psi_j(s, \tau_i(t, x); 0)$ we have

$$\begin{aligned}
\frac{d}{ds}\varphi_k(a_0, s; y) &= \frac{\partial \varphi_k}{\partial s}(a_0, s; y) + \frac{\partial \varphi_k}{\partial y}(a_0, s; y)\frac{\partial y}{\partial s} \\
&= -\frac{\partial \varphi_k}{\partial y}(a_0, s; y)\,\lambda_k(s, y) + \frac{\partial \varphi_k}{\partial y}(a_0, s; y)\,\mu_j(s, y) \\
&= \frac{\partial \varphi_k}{\partial y}(a_0, s; y)(\mu_j(s, y) - \lambda_k(s, y)) \neq 0.
\end{aligned}$$

Moreover,

$$\begin{aligned}
&\frac{d}{ds}\psi_k(a_0, \tau_k(s, y)); 0) \\
&\quad = \frac{\partial \psi_k}{\partial \tau_k}(a_0, \tau_k(s, y); 0)\left[\frac{\partial \tau_k}{\partial s}(s, y) + \frac{\partial \tau_k}{\partial y}(s, y)\frac{\partial \psi_j}{\partial s}\right] \\
&\quad = -\frac{\partial \psi_k}{\partial x}(a_0, \tau_k(s, y); 0)\,\mu_k(\tau_k(s, y); 0) \\
&\qquad \times\left[\frac{\partial \tau_k}{\partial s}(s, y) + \frac{\partial \tau_k}{\partial y}(s, y)\,\mu_j(s, y)\right]
\end{aligned}$$

and an elementary computation shows that

$$\frac{\partial \tau_k}{\partial s}(s, y) = \frac{\partial \varphi_k}{\partial y}(\tau_k(s, y), s; y)$$

and

$$\frac{\partial \tau_k}{\partial y}(s, y) = -\frac{1}{\lambda_k(s, y)} \frac{\partial \varphi_k}{\partial y}(\tau_k(s, y), s; y);$$

therefore the term between the brackets in the expression for $(d/ds)\,\psi_k(a_0, \tau_k(s, y); 0)$ is equal to

$$\frac{1}{\lambda_k(s, y)} \frac{\partial \varphi_k}{\partial y}(\tau_k(s, y), s; y)[\lambda_k(s, y) - \mu_j(s, y)]$$

which never vanishes. Using Lemma 3 we conclude the terms of type (II.5) define compact operators. In order to get the same conclusion for the terms (II.6) we first notice that they can be written in the form

$$\int_{a_0}^{\tau_i(t,x)} \tilde{c}_{jk}(s, y)\, \alpha_{ijk}(t, s, x)\, v_0^k(\psi_k(a_0, s; y))\, ds$$

and

$$\begin{aligned}
&\frac{d}{ds}\psi_k(a_0, s; y) \\
&\quad = \frac{\partial}{\partial s}\psi_k(a_0, s; y) + \frac{\partial \psi_k}{\partial y}(a_0, s; y)\frac{\partial \psi_j}{\partial s}(s, \tau_i(t, s); 0) \\
&\quad = \frac{\partial \psi_k}{\partial y}(a_0, s; y)[\mu_j(s, y) - \mu_k(s, y)];
\end{aligned}$$

if $j = k$ we see that the term (II.6) is absent because $\tilde{c}_{jj} \equiv 0$; moreover, by assumption v, if $j \neq k$ and $\mu_j = \mu_k$ somewhere then $\tilde{c}_{jk}$ is identically zero and this means that $(d/ds)\,\psi_k(a_0, s; y) \neq 0$ for the relevant terms and so, using Lemma 3 again, we conclude they define compact operators. The integrals

$$\int_{\tau_i(t,x)}^{t} \bar{u}^i(t, s, x)\, ds$$

can be treated in a completely way and then, by Lemma 2, we conclude $T(t, a) - T_3(t, a)$ is compact for any t in $[a, b]$ (in fact $\{(T(t, a) - T_3(t, a))(u_0, v_0, d_0),\ t \in J,\ |(u_0, v_0, d_0)| \leqslant 1\}$ is precompact). Finally, $T_3(t, a) - T_4(t, a)\Pi = T_3(t, a) - T_3(t, a)\Pi = T_3(t, a)(I - \Pi)$ is com-

pact because it has finite dimensional range (in fact, $\{(T_3(t, a) - T_4(t, a)\Pi)(u_0, v_0, d_0),\ t \in J,\ |(u_0, v_0, d_0)| \leqslant 1\}$ is precompact) and Theorem A is proved.

As a preparation for attacking the autonomous case we compute the resolvent operators $R(\lambda, A)$ and $R(\lambda, A_4)$ of the infinitesimal generators of $T(t)$ and $T_4(t)$. Recalling that

$$A: \mathscr{D}(A) \subset X_p \to X_p$$

is given by

$$A(u, v, d) = \left(-K(x) \frac{d}{dx} \begin{pmatrix} u \\ v \end{pmatrix} - C(x) \begin{pmatrix} u \\ v \end{pmatrix}, Fu(l) + Gv(l) \right)$$

with

$$\mathscr{D}(A) = \{(u, v, d) \in X_p : (u, v) \in (H_{1,p}[0, l])^n,\ u(0) = Ev(0)\ d = v(l) - Du(l)\}$$

we have to solve $(\lambda I - A)(u, v, d) = (f, g, b)$ or, equivalently

$$\frac{d}{dx} \begin{pmatrix} u \\ v \end{pmatrix} = -K^{-1}(x)(\lambda I + C(x)) \begin{pmatrix} u \\ v \end{pmatrix} + K^{-1}(x) \begin{pmatrix} f \\ g \end{pmatrix} \tag{II.7}$$

$$\lambda d = b + Fu(l) + Gv(l) \tag{II.8}$$

with the conditions

$$u(0) = Ev(0) \tag{II.9}$$

and

$$d = v(l) - Du(l). \tag{II.10}$$

Denoting by $Y(x, y, \lambda)$ the fundamental matrix of the system $(d/dx)\binom{u}{v} = -K^{-1}(x)(\lambda I + C(x))\binom{u}{v}$ from (II.7) and (II.9) we get

$$\begin{pmatrix} u(x) \\ v(x) \end{pmatrix} = Y(x, 0, \lambda) \begin{pmatrix} Ev(0) \\ v(0) \end{pmatrix} + \int_0^x Y(x, y, \lambda)\, K^{-1}(y) \begin{pmatrix} f(y) \\ g(y) \end{pmatrix} dy \tag{II.11}$$

and in order to satisfy (II.8) and (II.10), $v(0)$ has to solve

$$H(\lambda)\, v(0) = b + \int_0^l (\lambda D + F, G - \lambda I)\, Y(l, y, \lambda)\, K^{-1}(y) \begin{pmatrix} f(y) \\ g(t) \end{pmatrix} dy \tag{II.12}$$

where $H(\lambda) = -(\lambda D + F, G - \lambda I)\, X(l, 0, \lambda)\binom{E}{I}$. Defining $h(\lambda) = \det H(\lambda)$ we see that λ belongs to the resolvent set of A if and only if $h(\lambda) \neq 0$; in this case, $R(\lambda, A)(f, g, b) = (u, v, d)$, where (u, v) are given by (II.11) with $v(0)$

given by (II.12) and $d = v(l) - Du(l)$. In particular the spectrum $\sigma(A)$ of A consists of point spectrum only.

Let us denote by $Y_0(x, y, \lambda)$ the diagonal $n \times n$ matrix whose entries are

$$\exp\left(-\lambda \int_y^x \frac{ds}{\lambda_i(s)} - \int_y^x \frac{c_{ii}(s)}{\lambda_i(s)}\, ds \right)$$

for $i = 1, \ldots, N$, and the same expression for $i = N+1, \ldots, n$ with λ_i replaced by μ_i; writing $Y_0(x, y, \lambda)$ in blocks

$$\begin{bmatrix} Y_0^{11} & 0 \\ 0 & Y_0^{22} \end{bmatrix},$$

where Y_0^{11} is $N \times N$ and Y_0^{22} is $(n-N) \times (n-N)$ and defining $H_0(\lambda) = Y_0^{22}(l, 0, \lambda) - DY_0^{11}(l, 0, \lambda)E$, a short computation shows that a complex number λ belongs to the resolvent set of A_4 if and only if $h_4(\lambda) = \det H_0(\lambda) \neq 0$; in this case,

$$R(\lambda, A_4)\begin{pmatrix} f \\ g \end{pmatrix}(x)$$

$$= Y_0(x, 0, \lambda)\begin{pmatrix} Ev(0) \\ v(0) \end{pmatrix} + \int_0^x Y_0(x, y, \lambda)\, K^{-1}(y)\begin{pmatrix} f(y) \\ g(y) \end{pmatrix} dy$$

where

$$v(0) = H_0^{-1}(\lambda)\left[-\int_0^l Y_0^{22}(l, y, \lambda)\, g_1(y)\, dy + D\int_0^l Y_0^{11}(l, y, \lambda)\, f_1(y)\, dy \right],$$

with $\begin{pmatrix} f_1(y) \\ g_1(y) \end{pmatrix} = K^{-1}(y)\begin{pmatrix} f(y) \\ g(y) \end{pmatrix}$. Notice that $h_4(\lambda)$ is an exponential polynomial $\sum_{k=1}^{n_0} b_k e^{\lambda \omega_k}$.

The proof of next lemma uses the technique developed by Henry in [3].

LEMMA 5. *If α_0 is defined as in Theorem* B *then*

$$r(T_4(t)) = r_{\mathrm{ess}}(T_4(t)) = e^{\alpha_0 t}, \qquad t \geqslant 0.$$

Proof. First we show that $r(T_4(t)) \leqslant e^{\alpha_0 t}$, $t \geqslant 0$; this is equivalent to say that for any $\alpha > \alpha_0$ there is a constant K (depending on α) for which $|T_4(t)| \leqslant Ke^{\alpha t}$, $t \geqslant 0$; clearly it suffices to show that $|T_4(t)w| \leqslant Ke^{\alpha t}|w|$ for some K and any w in a dense subset of X_p. We know there is an $\beta > 0$ such that for any $w \in \mathscr{D}(A^2)$ we have

$$T_4(t)w = \frac{1}{2\pi i}\int_{\beta - i\infty}^{\beta + i\infty} e^{\lambda t} R(\lambda, A_4)w\, dy;$$

w will be taken smooth with compact support in $[0, l]$. The desired estimate will be obtained through this representation of $T_4(t)$. The expression for $R(\lambda, A_4)$ has two parts and we estimate the first only because the second is much simpler. First we show (Fig. 3) that the path of integration can be shifted to $\operatorname{Re}\lambda = \alpha$ by showing that the integrals on the upper and the lower boundaries of the region R in Fig. 3 go to zero. Notice that $|h_4(\lambda)|$ is bounded below from zero for $\alpha \leqslant \operatorname{Re}\lambda \leqslant \beta$ because such λ's are at a positive distance from the zeroes of $h_4(\lambda)$ is a finite exponential sum (see [3, Lemma 3.1]) and then a typical, term of $R(\lambda, A_4)w$ is the sum of terms of the form

$$a(x, \lambda) \int_0^l b(y)\, e^{-\lambda \int_0^y d\tau/\lambda_i(\tau)} f(y)\, dy,$$

where $a(x, \lambda)$ is bounded for $0 \leqslant x \leqslant l$ and $\alpha \leqslant \operatorname{Re}\lambda \leqslant \beta$, b is C^1 and f is a component of w; integrating by parts we get

$$\int_0^l b(y) f(y)\, e^{-\lambda \int_0^y ds/\lambda_1(s)}\, dy$$

$$= \frac{1}{\lambda} \int_0^l e^{-\lambda \int_0^y d\tau/\lambda_i(\tau)} \frac{d}{dy} (b(y) f(y) \lambda_i(y))\, dy$$

and this goes to zero (in $L_\infty([0, l])$) as $|\lambda| \to +\infty$, $\alpha \leqslant \operatorname{Re}\lambda \leqslant \beta$. We conclude

$$T_4(t)w = \frac{1}{2\pi i} \int_{\alpha - i\infty}^{\alpha + i\infty} e^{\lambda t} R(\lambda, A_4)\, d\lambda.$$

Next we notice that for $\lambda = \alpha + is$, $1/h_4(\lambda) = \sum_{k=1}^{\infty} a_k e^{\lambda \gamma_k}$, where γ_k is real and $\sum_{k=1}^{\infty} |a_k| r^{\alpha \gamma_k} < \infty$ (see [3, Lemma 3.4]) then we have to estimate terms of type

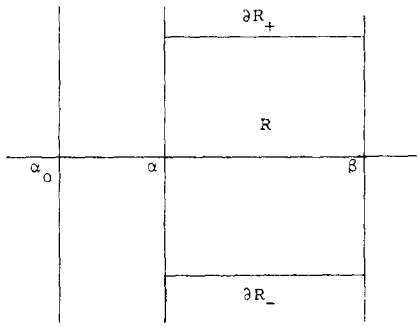


FIGURE 3

$$\lim_{r\to+\infty}\left[\int_{-r}^{r} e^{(\alpha+is)t}\left(\sum_{k=1}^{\infty} a_k e^{(\alpha+is)\gamma_k}\right)\right.$$

$$\times e^{-(\alpha+is)\int_0^x d\tau/\mu_j(\tau)-\int_0^x (c_{jj}(\tau)/\mu_j(\tau))\,d\tau}$$

$$\left.\times\int_0^l e^{-(\alpha+is)\int_y^l d\tau/\lambda_m(\tau)-\int_y^l (c_{mm}(\tau)/\lambda_m(\tau))d\tau} f(y)\,dy\right] ds$$

$$= \lim_{r\to+\infty}\sum_{k=1}^{\infty} a_k e^{\alpha\gamma_k} e^{\alpha t} b_1(x)$$

$$\int_0^l\left[\int_{-r}^{r} e^{is(t+\gamma_k-c(x)-d(y))}\,ds\right] b_2(y) f(y)\,dy$$

$$= \lim_{r\to+\infty} 2\sum_{k=1}^{\infty} a_k e^{\alpha\gamma_k} e^{\alpha t} b_1(x)$$

$$\times\int_0^l \frac{\operatorname{sen} r(t+\gamma_k-c(x)-d(y))}{t+\gamma_k-c(x)-d(y)}\, b_2(y) f(y)\,dy$$

where $b_1(x)$, $b_2(y)$ are C^1 functions, $c(x)=\int_0^x d\tau/\mu_j(\tau)$, and $d(y)=\int_y^l d\tau/\lambda_m(\tau)$; if we set $f=0$ outside $[0, l]$ and define the change of variables $z=d(y)$, $y=\eta(z)$ we get

$$\int_0^l \frac{\operatorname{sen} r(t+\gamma_k-c(x)-d(y)}{t+\gamma_k-c(x)-d(y)}\, b_2(y) f(y)\,dy$$

$$= \int_{-\infty}^{\infty} \frac{\operatorname{sen} r(t+\gamma_k-c(x)-z)}{t+\gamma_k-c(x)-z}\, b_3(z) f(\eta(z))\,dz,$$

where $b_3(z)=b_2(\eta(z))\,\eta'(z)$ and since the limit of this expression as $r\to+\infty$ is $\pi b_3(t+\gamma_k-c(x)) f(\eta(t+\gamma_k-c(x)))$ (by Fourier integral formula) we conclude the limit we are trying to estimate is equal to

$$2\pi\sum_{k=1}^{\infty} a_k e^{\alpha\gamma} e^{\alpha t} b_1(x)\, b_3(t+\gamma_k-c(x))\, f(\eta(t+\gamma_k-c(x));$$

the permutation of the limit in r with $\sum_{k=1}^{\infty}$ is legal because $\int_0^l$ above is uniformly bounded. Finally, the $L_p[0, l]$ norm of the summation above is bounded by $Me^{\alpha t}|f|_{L_p[0,l]}\sum_{k=1}^{\infty} a_k e^{\alpha\gamma_k}$ because η' and c' do not vanish and this gives the desired estimate of $|T_4(t)w|$. Furthermore, as a consequence of the almost periodicity of $h_4(\lambda)$ on vertical lines, we know there is a sequence λ_n such that $h_4(\lambda_n)=0$, $\operatorname{Re}\lambda_n\to\alpha_0$, and $|\lambda_n|\to+\infty$ (see [3, Lemma 3.2]) and this proves the lemma.

Proof of Theorem B. Assertions (a) and (b) have been already proved; assertion (c) follows from the explicit representation of $R(\lambda, A)$ and $R(\lambda, A_4)$. From Theorem A we know that

$$r_{\text{ess}}(T(t)) = r_{\text{ess}}(T_4(t)) = e^{\alpha_0 t}$$

and so, if z belongs to $\sigma(T(t))$ and $|z| > e^{\alpha_0 t}$ then z has to be in the point spectrum of $T(t)$; this means that z has to be equal to $e^{\lambda t}$ for some λ in the point spectrum of A and this proves (d). Assertion (e) follows from (d) and the previous comments. If (f) was false then the spectral radius of $T(t)$ restricted to $(I-P)X_p$ would be larger than $e^{\alpha t}$ (notice that $(I-P)X_p$ and PX_p are invariant under $T(t)$) and, since $r_{\text{ess}}(T(t)) = e^{\alpha_0 t} < e^{\alpha t}$, that would imply the existence of an eigenvalue of $T(t)$ larger than $e^{\alpha_0 t}$ corresponding to an eigenvector in $(I-P)X_p$ and this is impossible by assertion (e). Thus Theorem B is proved.

IV. Examples

As a first example we consider two vibrating strings with one end fixed and the other being connected through a mass m (which will be taken equal to one) subject to a force exerted by a spring and to a viscous damping (See Fig. 4). The equations of motion are

$$\frac{\partial U_1}{\partial t} = V_1, \qquad 0 < x < l$$

$$\frac{\partial V_1}{\partial t} = \frac{\partial}{\partial x}\left(a_1(t, x)\frac{\partial U_1}{\partial x}\right) - \beta_1(t, x)\, V_1 + b_1(t, x)\, U_1, \qquad 0 < x < l$$

$$\frac{\partial U_2}{\partial t} = V_2, \qquad l < x < 2l$$

$$\frac{\partial V_2}{\partial t} = \frac{\partial}{\partial x}\left(a_2(t, x)\frac{\partial U_2}{\partial x}\right) - \beta_2(t, x)\, V_2 + b_2(t, x)\, U_2, \qquad l < x < 2l$$

$$\frac{d}{dt} V_1(t, l) = -c(t)\, V_1(t, l) - k_1(t)\frac{\partial U_1}{\partial x}(t, l)$$

$$+ k_2(t)\frac{\partial U_2}{\partial x}(t, l) + k(t)\, U_1(t, l)$$

with boundary conditions $U_1(t, 0) = 0 = U_2(t, 2l)$, $U_1(t, l) = U_2(t, l)$. The coefficients are assumed to be as smooth as necessary and $a_i(t, x) > 0$, $i = 1, 2$. There is no loss of generality in assuming that the strings have the

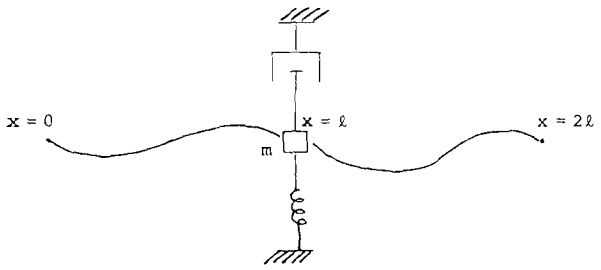


FIGURE 4

same length (if not we just rescale one of it). In order to put the system above in a convenient form we start by changing the interval $(l, 2l]$ into $[0, l]$ by the change $x \to -x + 2l$ and, keeping the same notation for the new coefficients, the new system is exactly the old one with k_2 replaced by $-k_2$ and $U_2(t, 2l) = 0$ by $U_2(t, 0) = 0$. For $1 \leqslant p < \infty$ we define

$$Y_p = \{(\varphi_1, \psi_1, \varphi_2, \psi_2, d) \in H_{1,p}[0, l] \times L_p[0, l] \times H_{1,p}[0, l] \times L_p[0, l] \times \mathbb{R}: \varphi_1(0) = 0, \varphi_2(0) = 0, \varphi_1(l) = \varphi_2(l)\}$$

and $A(t)$: $\mathscr{D}(A(t)) \subset Y_p \to Y_p$ by

$$A(t)(\varphi_1, \psi_1, \varphi_2, \psi_2, d) = \left(\psi_1, \frac{d}{dx}\left(a_1(t, x)\frac{d\varphi_1}{\partial x}\right) - \beta_1(t, x)\psi_1 + b_1(t, x)\varphi_1, \right.$$

$$\psi_2, \frac{d}{dx}\left(a_2(t, x)\frac{d\varphi_2}{dx}\right) - \beta_2(t, x)\psi_2 + b_2(t, x)\varphi_2, -c(t)\psi_1(l)$$

$$\left. -k_1(t)\frac{d}{dx}\varphi_1(l) - k_2(t)\frac{d}{dx}\varphi_2(l) + k(t)\varphi_1(l)\right),$$

with

$$\mathscr{D}(A(t)) = \{(\varphi_1, \psi_1, \varphi_2, \psi_2, d) \in Y_p: \varphi_1, \varphi_2 \in H_{2,p}, \psi_1, \psi_2 \in H_{1,p}, d = \psi_1(l), \psi_1(0) = 0 = \psi_2(0), \psi_1(l) = \psi_2(l)\}.$$

With that notation $A(t)$ generates an evolution operator on Y_p and the problem is to study its essential spectral radius. First of all we notice that $(\varphi_1, \psi_1, \varphi_2, \psi_2, d) \in Y_p \to (0, b_1(t, x)\varphi_1, 0, b_2(t, x), \quad k(t)\varphi_1(l)) \in Y_p$ is a compact operator and then, by Lemma 3, we can drop those terms without

changing the essential spectral radius. Next we introduce the Riemann invariants

$$u_1 = V_1 + \alpha_1 \frac{\partial U_1}{\partial x}, \qquad v_1 = V_1 - \alpha_1 \frac{\partial U_1}{\partial x}, \qquad u_2 = V_2 + \alpha_2 \frac{\partial U_2}{\partial x},$$

$v_2 = V_2 - \alpha_2(\partial U_2/\partial x)$, where $\alpha_1 = \sqrt{a_1}$, $\alpha_2 = \sqrt{a_2}$ and our system becomes

$$\frac{\partial u_i}{\partial t} + \alpha_i \frac{\partial u_i}{\partial x} + \frac{1}{2}\left(\beta_i + \frac{\partial \alpha_i}{\partial x} - \frac{1}{\alpha_i}\frac{\partial \alpha_i}{\partial t}\right) u_i$$

$$+ \frac{1}{2}\left(\beta_i - \frac{\partial \alpha_i}{\partial x} + \frac{1}{\alpha_i}\frac{\partial \alpha_i}{\partial t}\right) v_i = 0$$

$$\frac{\partial v_i}{\partial t} - \alpha_i \frac{\partial v_i}{\partial x} + \frac{1}{2}\left(\beta_i + \frac{\partial \alpha_i}{\partial x} + \frac{1}{\alpha_i}\frac{\partial \alpha_i}{\partial t}\right) u_i$$

$$+ \frac{1}{2}\left(\beta_i - \frac{\partial \alpha_i}{\partial x} - \frac{1}{\alpha_i}\frac{\partial \alpha_i}{\partial t}\right) v_i = 0 \qquad i = 1, 2$$

$$\frac{d}{dt}\left[v_i(t, l) + u_1(t, l)\right]$$

$$= -c(t)(v_1(t, l) + u_1(t, l))$$

$$- \frac{k_1(t)}{\alpha_1(t, l)}(u_1(t, l) - v_1(t, l)) - \frac{k_2(t)}{\alpha_2(t, l)}(u_2(t, l) - v_2(t, l))$$

and boundary conditions $u_1(t, 0) + v_1(t, 0) = 0$, $u_2(t, 0) + v_2(t, 0) = 0$, $u_1(t, l) + v_1(t, l) = u_2(t, l) + v_2(t, l)$. This system is not exactly of the the type we have worked with because at $x = l$ there is a boundary condition (instead of a time derivative) but, as we have pointed out, it can be treated by the same method and the conclusion is that the evolution operator $T_0(t, s)$ generated by the reduced system:

$$\frac{\partial u_i}{\partial t} + \alpha_i \frac{\partial u_i}{\partial x} + \frac{1}{2}\left(\beta_i + \frac{\partial \alpha_i}{\partial x} - \frac{1}{\alpha_i}\frac{\partial \alpha_i}{\partial t}\right) u_i = 0$$

$$\frac{\partial v_i}{\partial t} - \alpha_i \frac{\partial v_i}{\partial x} + \frac{1}{2}\left(\beta_i + \frac{\partial \alpha_i}{\partial x} + \frac{1}{\alpha_i}\frac{\partial \alpha_i}{\partial t}\right) v_i = 0, \qquad i = 1, 2$$

with boundary conditions $u_i(t, 0) + v_i(t, 0) = 0$, $u_i(t, l) + v_i(t, l) = 0$, $i = 1, 2$, and the evolution operator generated by the complete system have the same essential spectral radius. The computation of $r_{\text{ess}}(T_0(\tau, 0))$ in the case of periodic coefficients seems to be difficult in the general case; in the case β_i are constant, $i = 1, 2$, that has been carried out in [11]; if α_i and β_i are

independent of t (the autonomous case) then a short computation shows that the characteristic roots of the reduced equation satisfy

$$2\lambda \int_0^l \frac{ds}{\alpha_1(s)} = -\int_0^l \frac{\beta_1(s)}{\alpha_1(s)}\, ds + 2n\pi\sqrt{-1}$$

and according to TheoremII(B), $r_{\text{ess}}(T(t)) = e^{-\alpha_0 t}$,

$$\alpha_0 = \min_{i=1,2} \frac{1}{2}\left(\int_0^l \frac{\beta_i(s)}{\alpha_i(s)}\, ds \Big/ \int_0^l \frac{ds}{\alpha_i(s)}\right).$$

As a second example we consider the system:

$$U_{tt} = a_1 U_{xx} + b_1 V_{xx} + C_1 U_t + d_1 V_t + e_1 U_x + f_1 V_x + g_1 U + h_1 V$$
$$V_{tt} = a_2 U_{xx} + b_2 V_{xx} + C_2 U_t + d_2 V_t + e_2 U_x + f_2 V_x + f_2 U + h_2 V$$

$0 < x < l$, with boundary conditions $U(t, 0) = V(t, 0) = U(t, l) = V(t, l) = 0$, where the coefficients are smooth functions of (t, x); we assume the matrix $\left[\begin{smallmatrix} a_1 & b_1 \\ a_2 & b_2 \end{smallmatrix}\right]$ is symmetric positive definite with distinct eigenvalues $\varepsilon_i(t, x)$, $i = 1, 2$. Probably the system above has to do with the motion of a string with fixed ends and which is not restricted to a plane; $(U(t, x), V(t, x))$ represents the displacement from the equilibrium. As before the terms involving U and V can be dropped without changing the essential spectral radius. We introduce the Riemann invariants

$$\begin{bmatrix} u_1 \\ u_2 \\ v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 1 & \alpha_1 & \beta_1 & \gamma_1 \\ 1 & \alpha_2\alpha\beta_2 & & \gamma_2 \\ 1 & \alpha_3 & \beta_3 & \gamma_3 \\ 1 & \alpha_4 & \beta_4 & \gamma_4 \end{bmatrix} \begin{bmatrix} U_t \\ V_t \\ U_x \\ V_x \end{bmatrix}$$

where $\alpha_i = (\varepsilon_i^2 - \alpha_1)/a_2$, $\beta_1 = -\varepsilon_1$, $\beta_2 = -\varepsilon_2$, $\beta_3 = \varepsilon_1$, $\beta_4 = \varepsilon_4$, $\gamma_i = \beta_i\alpha_i$, and our system becomes

$$\frac{\partial}{\partial t}\begin{pmatrix} u \\ v \end{pmatrix} + K(t, x)\frac{\partial}{\partial x}\begin{pmatrix} u \\ v \end{pmatrix} + C(t, x)\begin{pmatrix} u \\ v \end{pmatrix} = 0$$

where $K(t, x) = \operatorname{diag}(\varepsilon_1(t, x), \varepsilon_2(t, x), -\varepsilon_1(t, x), -\varepsilon_2(t, x))$, and $C(t, x)$ can be obtained explicitly from the coefficients of the system above. The boundary conditions are obtained by replacing U_t and V_t by their expressions in terms of u_1, u_2, v_1, and v_2 in $U_t(t, 0) = U_t(t, l) = V_t(t, 0) = V_t(t, 0) = 0$ and Theorem II(A) and II(B) apply as in the first example.

As the final example we consider the system:

$$L\frac{\partial i}{\partial t} = -\frac{\partial v}{\partial x} - Ri \qquad 0 < x < l,$$

$$C\frac{\partial v}{\partial t} = -\frac{\partial i}{\partial x} - Gv \qquad 0 < x < l,$$

$$C_1 \frac{d}{dt} v(l, t) = i(l, t) + R_1 v(l, t)$$

$$v(0, t) + R_0 i(0, t) = 0$$

which is the linearized version studied by Brayton and Miranker [15]. Introducing the change of variables

$$\begin{bmatrix} U \\ V \end{bmatrix} = \frac{1}{2} \begin{bmatrix} \frac{1}{z_0} & 1 \\ \frac{1}{z_0} & -1 \end{bmatrix} \begin{bmatrix} v \\ i \end{bmatrix}, \qquad z_0 = \sqrt{\frac{E}{C}}$$

we obtain

$$\frac{\partial}{\partial t}\begin{bmatrix} U \\ V \end{bmatrix} + \begin{bmatrix} \lambda_1 & 0 \\ 0 & \mu_1 \end{bmatrix} \frac{\partial}{\partial x}\begin{bmatrix} U \\ V \end{bmatrix} + \begin{bmatrix} a & b \\ b & a \end{bmatrix}\begin{bmatrix} U \\ V \end{bmatrix} = 0$$

$$\frac{d}{dt}[V(l, t) + U(l, t)] = fU(l, t) + gV(l, t)$$

$$U(0, t) + qV(0, t) = 0$$

where $\lambda_1 = -\mu_1 = 1/\sqrt{LC}$, $2LCa = RC + LG$,

$$-2LCb = RC - LG, \qquad q = \frac{Z_0 - R_0}{Z_0 + R_0}.$$

A short computation shows that

$$h_4(\lambda) = \exp\left[\frac{l}{\sqrt{LC}}(\lambda + a)\right] - q \exp\left[-\frac{l}{\sqrt{LC}}(\lambda + a)\right]$$

and then the solutions of $h_4(\lambda) = 0$ are

$$\lambda_j = -a + \frac{c}{2l} \ln |q| + \frac{jc\pi i}{2l}, \qquad j = 0, \pm 1, \pm 2, \ldots$$

and so

$$\alpha_0 = -\frac{(RG + LG)}{2LC} + \frac{1}{2l}\sqrt{\frac{L}{C}} \ln \left| \frac{Z_0 - R_0}{Z_0 + R_0} \right|.$$

According to Theorem B the essential radius generated by the system above is $\exp \alpha_0 t$.

Parts of this work are contained in the theses of the first two authors written under the supervision of the third.

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